

Linking multiple CS548, CS328A or CS448 units with common trigger using CS1025 CS1020 CS1021 Cables

Summary

This instruction shows how to connect two to four CS328A or CS320A, or two CS548 oscilloscopes to make them into one 4-8 channel oscilloscope. This example used Cleverscope4 x64.exe version 5.1.5. (See Help/About Cleverscope).

CS1025 Link Cable

Connect from 2-4 CS328A's to the CS1024 by plugging the mini Din cable into the CS328A Link Ports. The Link Port Master cable plugs into the CS328A which will be the trigger master. Only the trigger master unit can do the triggering. Cables are equal length to ensure time alignment between the units.



CS1020 Link Cable

This cable is used to share trigger between two CS328A's. Plug each end of the link cable into a CS328A link port. Either unit can do the triggering.



CS1021 Link Cable

This cable is used to share trigger and sampling clock between two CS448 or CS548. Plug one end into a Link Out port, and the other end into a Link In Port.



Connect power, USB and probes

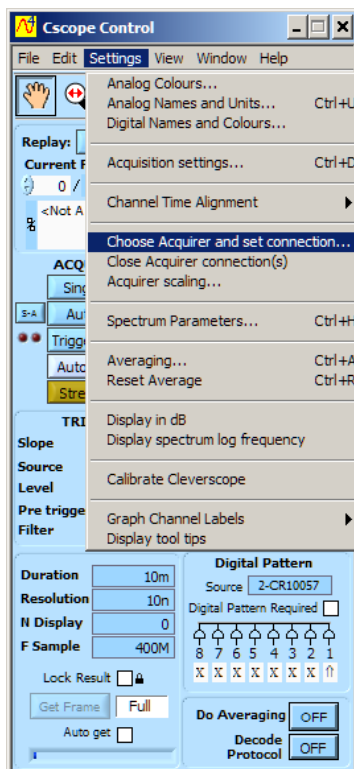
Connect the power supply to the Power In sockets, the USB cables, and any probes you want to use.

Waiting for Startup

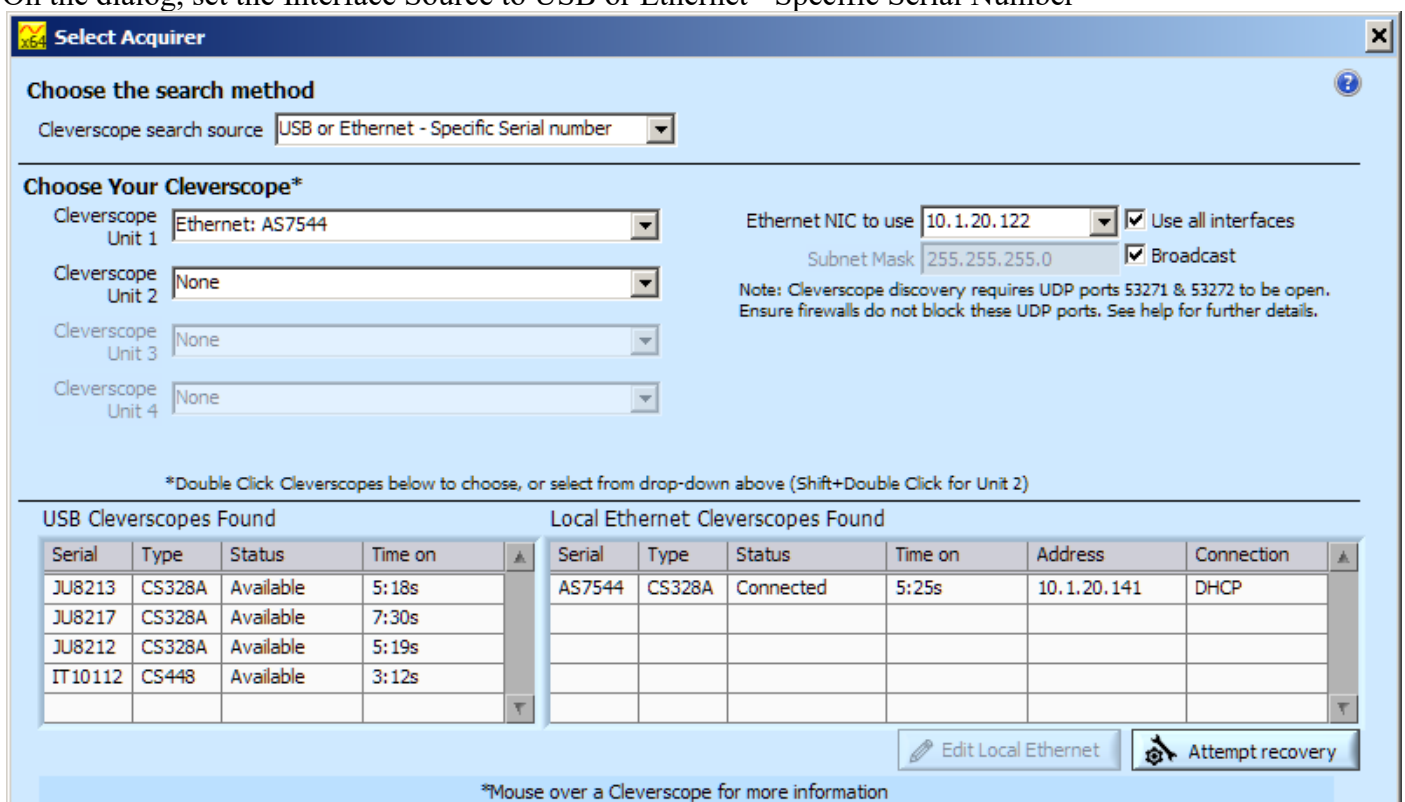
After plugging in the power, the green Power Leds will flash on the CS448. During this period, the unit is performing self test, and programming the isolated channels and isolated signal generator. The CS328A just outputs a brief flash. When the green led changes to continuously on, the unit is ready to use. A unit that is flashing will not capture signals.

First Time Use as an 8 channel scope

Open the Cleverscope4 x64 application, and navigate to the Settings/Choose Acquirer and set connection dialog:



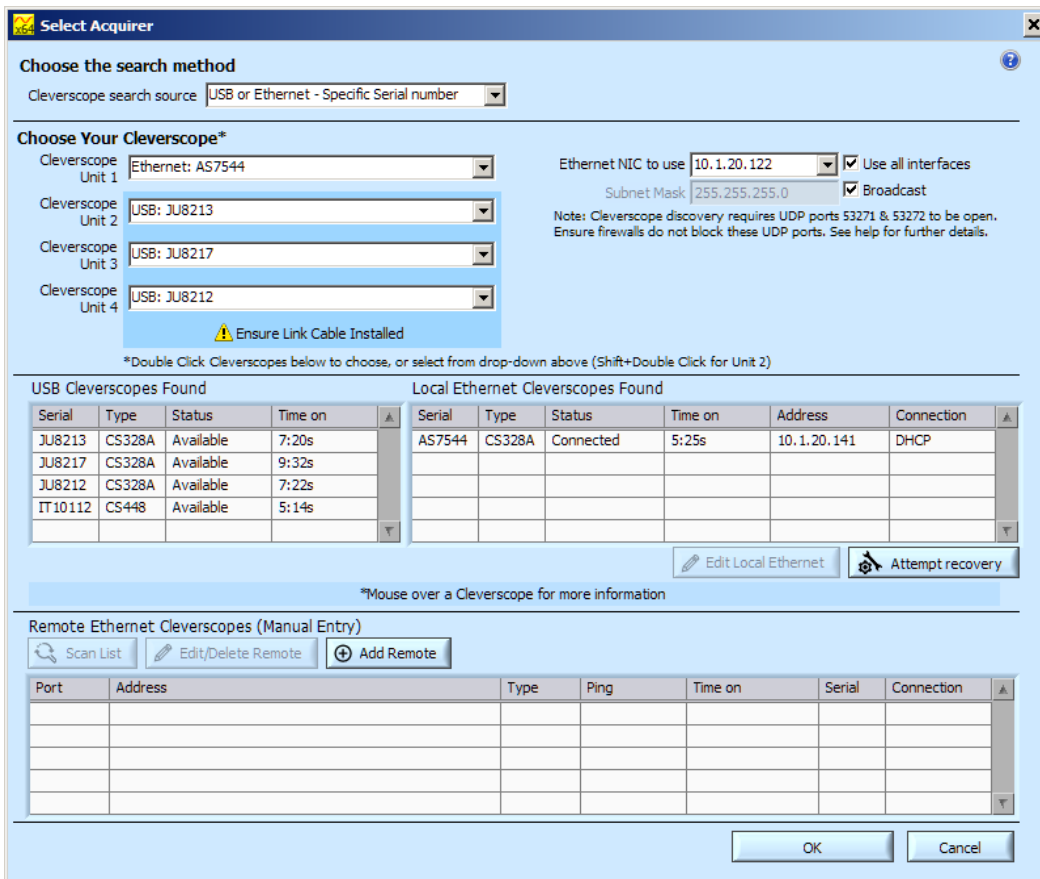
On the dialog, set the Interface Source to USB or Ethernet - Specific Serial Number



Here there are 4 x CS328A units available, one is a local ethernet, and the other three are USB. There is also a CS44 connected.

When using the CS1025, only unit 1 can be used as a trigger unit. When using a CS1020 or CS1020 any unit can be the trigger unit.

Select the Units to be connected. Note that Unit 1 becomes channels A..B (or A..D), unit 2 becomes channels C..D (or E..H) etc.



Select Acquirer

Choose the search method
 Cleverscope search source: **USB or Ethernet - Specific Serial number**

Choose Your Cleverscope*

Cleverscope Unit 1: **Ethernet: AS7544**
 Cleverscope Unit 2: **USB: JU8213**
 Cleverscope Unit 3: **USB: JU8217**
 Cleverscope Unit 4: **USB: JU8212**

Ethernet NIC to use: **10.1.20.122** ☒ Use all interfaces
 Subnet Mask: **255.255.255.0** ☒ Broadcast

Note: Cleverscope discovery requires UDP ports 53271 & 53272 to be open. Ensure firewalls do not block these UDP ports. See help for further details.

USB Cleverscopes Found

Serial	Type	Status	Time on
JU8213	CS328A	Available	7:20s
JU8217	CS328A	Available	9:32s
JU8212	CS328A	Available	7:22s
IT10112	CS448	Available	5:14s

Local Ethernet Cleverscopes Found

Serial	Type	Status	Time on	Address	Connection
AS7544	CS328A	Connected	5:25s	10.1.20.141	DHCP

Remote Ethernet Cleverscopes (Manual Entry)

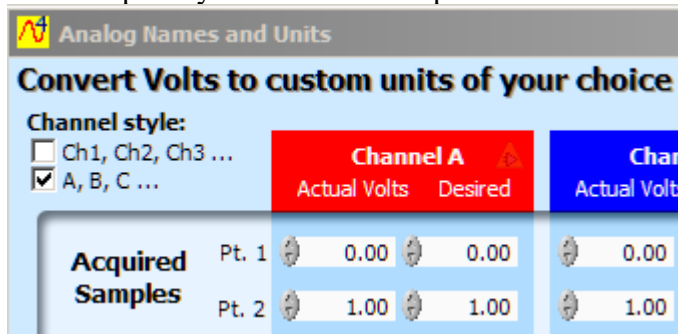
Scan List Edit/Delete Remote Add Remote

Port	Address	Type	Ping	Time on	Serial	Connection

OK Cancel

Click OK to confirm selection. The units will become connected.

You can choose to have the channels labelled A-H or, 1-8, by using Settings/Analog Names and Units. On the top left you will find the option to choose a Channel Style of Ch1, Ch2.. or A,B,C..



Analog Names and Units

Convert Volts to custom units of your choice

Channel style:
☐ Ch1, Ch2, Ch3 ...
☒ A, B, C ...

		Channel A		Channel B
		Actual Volts	Desired	Actual Volts
Acquired Samples	Pt. 1	0.00	0.00	0.00
	Pt. 2	1.00	1.00	1.00

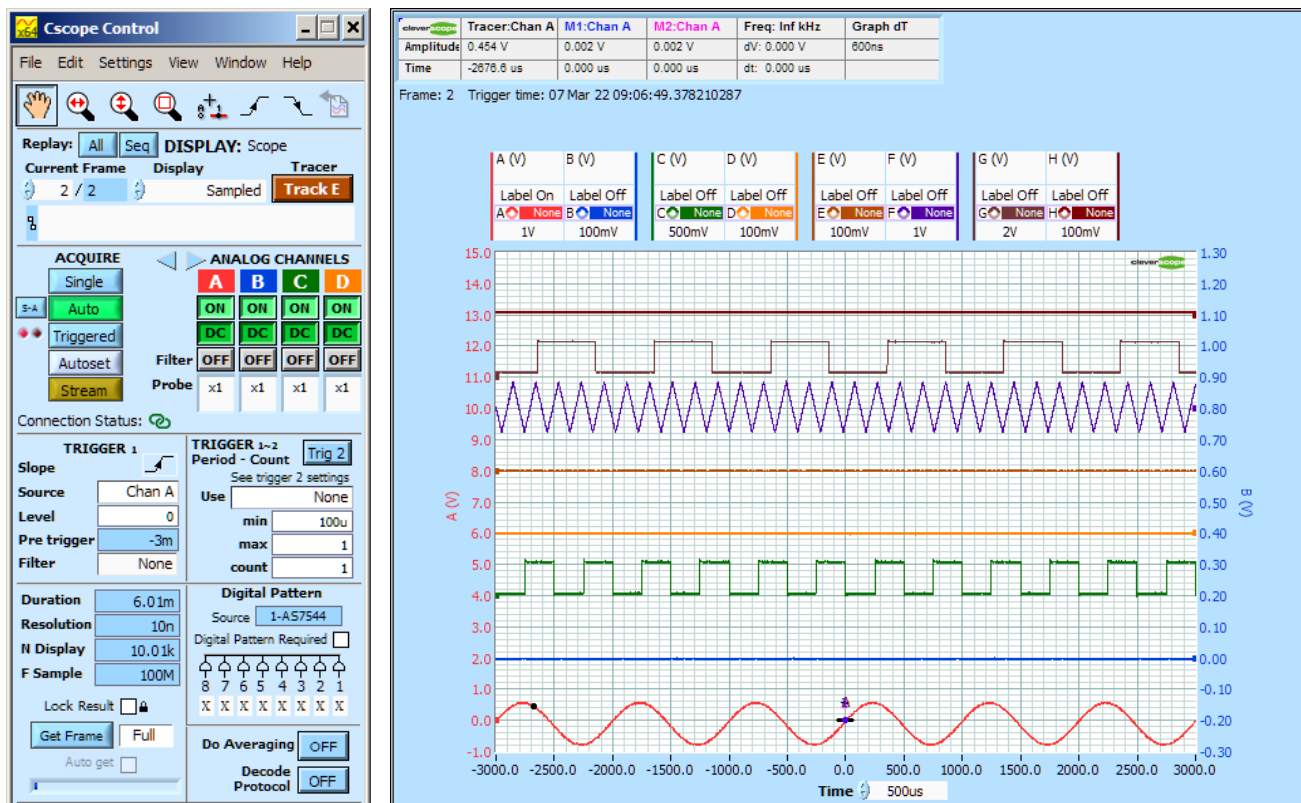
Note that the existing channel names will not be over-ridden. You can set them to the default by clicking the **Use default names and units** button. You can also just over type the Channel name in the Channel control above the scope display graph.

Setup triggering as needed over the eight channels, noting that with the CS1025 only Channels A or B can be trigger channels.

Start Capturing

Click on Auto to start capturing. You should see signals (or flat lines if there are no signals). You are now ready to start.

Here with 4 units capturing, Channel A is the trigger, with various signals on four of the channels:



Connecting again in the Future

Close the application when you no longer want to use the oscilloscope. Disconnect power to the scopes.

To start again, power the oscilloscopes, and wait until the green led is no longer flashing. Start the Cleverscope4 application. It will remember your last setup. Click Auto to start capturing.

If you have questions email support@cleverscope.com